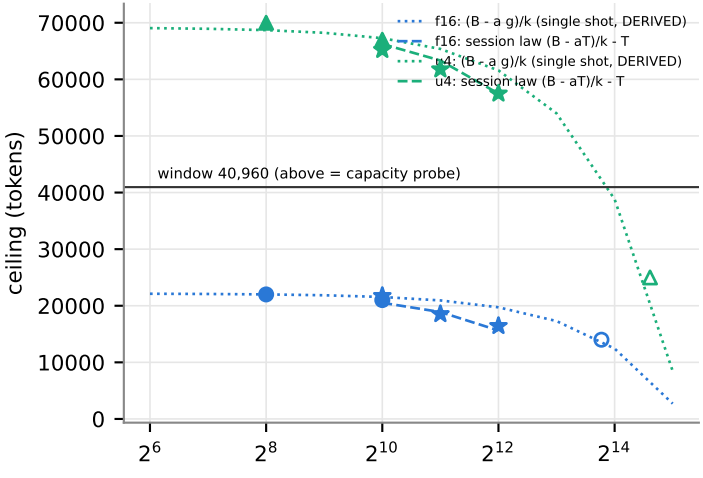
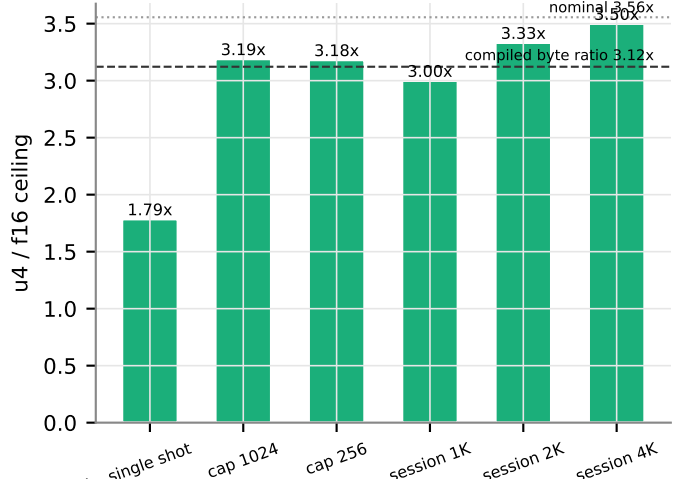

Ceiling vs prefill granularity (XPS)
filled = caps, hollow = single shot, stars = sessions
(last held turn); lines DERIVED, in-sample B
f16: (B - a g)/k (single shot, DERIVED)
f16: session law (B - aT)/k - T
u4: (B - a g)/k (single shot, DERIVED)
u4: session law (B - aT)/k - T
window 40,960 (above = capacity probe)
ceiling (tokens)
70000
60000
50000
40000
30000
20000
10000
0
$2^6$
$2^8$
$2^{10}$
$2^{12}$
$2^{14}$
prefill granularity g (tokens per step: cap, turn size, or the whole prompt)


KV precision x granularity:
u4 buys 1.79x unchunked, ~3.2x chunked
nominal 3.56x
compiled byte ratio 3.12x
u4 / f16 ceiling
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.79x
3.19x
3.18x
3.00x
3.33x
3.50x
single shot
cap 1024
cap 256
session 1K
session 2K
session 4K